

PS3/BS3 Functional Evidence Analysis

ACMG/AMP Variant Classification Report (PS3/BS3 criteria)

1. Variant Information

GENOMIC COORDINATES

2:162277580

ALLELES

C → A

GENE SYMBOL

IFIH1

RSID

rs35744605

HGVSC

ENST00000649979.2:c.1879G>T

HGVSP

ENSP00000497271.1:p.Glu627Ter

2. Literature Summary

73

Candidate Papers

19

Functional Papers

18

Experiments

3. Functional Experiments

Experiment 1 (Title: Loss-of-function mutations E6 27X and I923V of IFIH1 are associated with lower poly(I:C)-induced interferon- β production in peripheral blood mononuclear cells of type 1 diabetes patients. PMID: 20736039) **supports_pathogenic**

Justification of Paper Inclusion: The abstract discusses specific variants of the IFIH1 gene (E627X and I923V) and their association with decreased poly(I:C)-induced interferon- β production in peripheral blood mononuclear cells. This indicates a wet-lab functional assay outcome related to the variants, fulfilling the criteria for functional evidence.

Assay Type: ELISA

System: Human PBMCs

Readout: IFN- β secretion levels (pg/ml)

Effect Direction: loss_of_function

Magnitude & Stats: 620 ± 336 (infected T1D patients) vs 1360 ± 422 (noninfected controls), $p < 0.0001$

Controls & Quality: PBMCs from healthy controls and T1D patients used for comparisons; statistical significance assessed using Mann–Whitney U test.

Authors' Conclusion: E627X mutation results in reduced IFN- β production in response to poly(I:C) stimulation in T1D patients.

Experiment 2 (Title: Reduced expression of IFIH1 is protective for type 1 diabetes. PMID: 20844740) **supports_pathogenic**

Justification of Paper Inclusion: The abstract discusses the rare alleles, including rs35744605 (Glu627X), and presents experimental evidence of reduced IFIH1 expression through allele-specific expression analysis and quantitative PCR assays. This indicates a functional impact related to the variants, fulfilling the criteria for functional evidence.

Assay Type: Quantitative PCR assay

System: Human PBMCs

Readout: Level of IFIH1 mRNA detected

Effect Direction: loss_of_function

Magnitude & Stats: No significant change in IFIH1 mRNA levels detected in resting immune cells.

Controls & Quality: Control individuals without variants were used as comparators.

Authors' Conclusion: The Glu627X mutation leads to reduced expression of IFIH1 protein in heterozygous individuals.

Experiment 3 (Title: Admixture mapping in lupus identifies multiple functional variants within IFIH1 associated with apoptosis, inflammation, and autoantibody production. PMID: 23441136)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses multiple variants in the IFIH1 gene, including missense variants, and provides experimental validation of the intronic SNP's functional impact on protein binding and transcriptional activity. This indicates a clear link between the genetic variants and their functional consequences in the context of apoptosis and inflammation.

Assay Type: apoptosis assay

System: K562 leukemia cell line

Readout: Percentage of apoptotic cells by FACS

Effect Direction: loss_of_function

Magnitude & Stats: 14.6% average increase between risk allele 'A' versus protective allele 'G', $P = 0.014$

Controls & Quality: Cells transfected with non-conditional vectors did not induce apoptosis

Authors' Conclusion: Risk 'A' allele for rs10930046 significantly increases apoptosis compared to protective 'G' allele.

Experiment 4 (Title: Gain-of-function mutations in IFIH1 cause a spectrum of human disease phenotypes associated with upregulated type I interferon signaling. PMID: 24686847) supports_pathogenic

Justification of Paper Inclusion: The abstract discusses heterozygous mutations in IFIH1 and describes cellular and biochemical assays that demonstrate these mutations confer gain of function, leading to increased interferon signaling. This indicates a clear functional impact of the variant on gene product function.

Assay Type: cellular and biochemical assays

System: not specified

Readout: interferon signaling

Effect Direction: gain_of_function

Magnitude & Stats: not specified

Controls & Quality: not specified

Authors' Conclusion: mutant IFIH1 binds RNA more avidly, leading to increased baseline and ligand-induced interferon signaling

Experiment 5 (Title: The A946T variant of the RNA sensor IFIH1 mediates an interferon program that limits viral infection but increases the risk for autoimmunity. PMID: 28553952)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the IFIH1T946 variant and its impact on type I interferon production in both human cells and a knock-in mouse model. It provides evidence of functional outcomes, such as heightened interferon production and enhanced survival in viral challenges, linking the variant to specific functional assays.

Assay Type: cytokine production

System: human peripheral blood mononuclear cells (PBMCs)

Readout: production of type I interferons

Effect Direction: strong_loss_of_function

Magnitude & Stats: heightened basal and ligand-triggered production

Controls & Quality: compared to cells expressing wildtype IFIH1

Authors' Conclusion: The A946T variant mediates an interferon program that limits viral infection but increases the risk for autoimmunity.

Experiment 6 (Title: The A946T variant of the RNA sensor IFIH1 mediates an interferon program that limits viral infection but increases the risk for autoimmunity. PMID: 28553952)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the IFIH1T946 variant and its impact on type I interferon production in both human cells and a knock-in mouse model. It provides evidence of functional outcomes, such as heightened interferon production and enhanced survival in viral challenges, linking the variant to specific functional assays.

Assay Type: viral challenge

System: mice with a knock-in mutation encoding IFIH1 T946

Readout: survival after lethal viral challenge

Effect Direction: strong_loss_of_function

Magnitude & Stats: enhanced basal expression of type I interferons

Controls & Quality: compared to wildtype mice

Authors' Conclusion: Mice with IFIH1 T946 displayed enhanced survival against viral infection.

Experiment 7 (Title: The A946T variant of the RNA sensor IFIH1 mediates an interferon program that limits viral infection but increases the risk for autoimmunity. PMID: 28553952)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the IFIH1T946 variant and its impact on type I interferon production in both human cells and a knock-in mouse model. It provides evidence of functional outcomes, such as heightened interferon production and enhanced survival in viral challenges, linking the variant to specific functional assays.

Assay Type: autoimmunity models

System: mice with a knock-in mutation encoding IFIH1 T946

Readout: penetrance in autoimmune models

Effect Direction: strong_loss_of_function

Magnitude & Stats: increased penetrance in autoimmune models

Controls & Quality: compared to wildtype mice

Authors' Conclusion: The variant increases the risk of autoimmunity.

Experiment 8 (Title: Recurrent rhinovirus infections in a child with inherited MDA5 deficiency. PMID: 28606988) **supports_pathogenic**

Justification of Paper Inclusion: The abstract discusses a homozygous missense mutation in IFIH1 that leads to impaired recognition of dsRNA and reduced IFN induction. It provides evidence from functional assays showing that mutant MDA5 fails to drive luciferase activity and that cells expressing mutant MDA5 have increased replication of HRV, indicating a clear functional impact of the variant.

Assay Type: Luciferase reporter assay

System: 293T cells

Readout: Luciferase activity driven by IFNB1, ISRE, or NF-κB promoter

Effect Direction: loss_of_function

Magnitude & Stats: K365E mutation showed minimal activity above nontransfected cells.

Controls & Quality: Comparative controls with wild-type MDA5.

Authors' Conclusion: K365E failed to drive luciferase activity from IFN-stimulated response element (ISRE)- and NF-κB-driven promoters, indicating impaired function.

Experiment 9 (Title: Exome-wide association study reveals novel psoriasis susceptibility locus at TNFSF15 and rare protective alleles in genes contributing to type I IFN signalling. PMID: 28973304)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses protective rare variants in IFIH1, including the specific variant p.Glu627Ter, and mentions that these variants disrupt conserved protein domains, suggesting a functional impact on protein function. This indicates that there is experimental evidence linking the variant to a functional outcome in the context of type I IFN signaling.

Assay Type: gene-based association test

System: exome array

Readout: association P-value

Effect Direction: loss_of_function

Magnitude & Stats: pburden= 2.53×10^{-7} ; OR=0.707

Controls & Quality: Standard quality control procedures applied

Authors' Conclusion: The variants have significant protective associations against psoriasis risk.

Experiment 10 (Title: Recurrent and Prolonged Infections in a Child with a Homozygous IFIH1 Nonsense Mutation. PMID: 29018476)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses a homozygous nonsense mutation in IFIH1 and reports Western blot analysis showing absence of wild type MDA5/IFIH1 protein, indicating a functional consequence of the variant. This aligns with the criteria for functional evidence as it links the variant to a measurable outcome in protein expression.

Assay Type: Western blot analysis

System: cultured fibroblasts of the patient

Readout: absence of wild type MDA5/IFIH1

Effect Direction: strong_loss_of_function

Magnitude & Stats: compatible with NMD

Controls & Quality: Not specified in the text

Authors' Conclusion: Homozygous loss of IFIH1 defines a novel Mendelian immunodeficiency disorder that increases susceptibility to severe viral infections.

Experiment 11 (Title: Breaching Self-Tolerance to Alu Duplex RNA Underlies MDA5-Mediated Inflammation. PMID: 29395326)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses variants of MDA5, including the specific variant of interest, and describes functional outcomes related to the variant's ability to recognize dsRNA and form signaling-competent filaments. This indicates a wet-lab functional assay with a clear outcome related to the variant's effect on protein function.

Assay Type: IRF3 dimerization assay

System: Cell-free system using cytosolic extract from 293T cells

Readout: IRF3 dimerization as measured by native gel electrophoresis

Effect Direction: loss_of_function

Magnitude & Stats: IRF3 dimerization stimulated significantly in the presence of dsRNA

Controls & Quality: K63-linked polyubiquitin was used to stabilize oligomers; controls involving GST-2CARD

Authors' Conclusion: G495R variant enhances IRF3 dimerization in response to specific dsRNAs, indicating aberrant signaling

Experiment 12 (Title: IFIH1 loss-of-function variants contribute to very early-onset inflammatory bowel disease. PMID: 34185153)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses loss-of-function variants in the IFIH1 gene and describes the use of luciferase reporter assays to assess MDA5 activity, indicating a wet-lab functional assay. It also mentions the impact of these variants on MDA5 function and their association with Very Early Onset Inflammatory Bowel Disease, providing a clear outcome related to the variants.

Assay Type: Luciferase reporter assay

System: 293T human cells

Readout: Normalized luciferase activity

Effect Direction: loss_of_function

Magnitude & Stats: Robust activity induced by wild-type MDA5; p.Glu627Ter failed to activate the IFNB1 promoter.

Controls & Quality: Empty vector and wild-type controls included during transfection

Authors' Conclusion: The p.Glu627Ter variant results in complete loss of MDA5 function.

Experiment 13 (Title: Human enteric viruses autonomously shape inflammatory bowel disease phenotype through divergent innate immunomodulation. PMID: 35394816) **supports_benign**

Justification of Paper Inclusion: The abstract mentions 'loss-of-function mutations within virus sensor MDA5' and discusses the detrimental consequences for IBD patient-derived intestinal epithelial cells, indicating a functional impact related to the genetic variant. This suggests that the variant affects the function of the gene product in a disease-relevant context.

Assay Type: ELISA

System: primary human macrophages

Readout: IL-10 levels

Effect Direction: no_effect_vs_wildtype

Magnitude & Stats: IL-10 levels significantly higher in non-IBD virome treatment

Controls & Quality: Mock control using unexposed macrophages confirmed baseline levels of IL-10.

Authors' Conclusion: Healthy enteric viromes maintain intestinal immunity by promoting anti-inflammatory responses.

Experiment 14 (Title: Human enteric viruses autonomously shape inflammatory bowel disease phenotype through divergent innate immunomodulation. PMID: 35394816) supports_pathogenic

Justification of Paper Inclusion: The abstract mentions 'loss-of-function mutations within virus sensor MDA5' and discusses the detrimental consequences for IBD patient-derived intestinal epithelial cells, indicating a functional impact related to the genetic variant. This suggests that the variant affects the function of the gene product in a disease-relevant context.

Assay Type: Transfection

System: HEK293T cells

Readout: IFN- β luciferase activity

Effect Direction: loss_of_function

Magnitude & Stats: MDA5E627X exhibited significantly reduced IFN- β activity compared to wild-type.

Controls & Quality: Comparative analysis with wild-type MDA5 confirmed loss of function in mutant forms.

Authors' Conclusion: Loss-of-function variants in MDA5 lead to impaired antiviral responses to enteric viromes.

Experiment 15 (Title: Age-dependent effect of the IFIH1/MDA5 gene variants on the risk of critical COVID-19. PMID: 36434151)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses the IFIH1 variant rs35744605 (p.Glu627Stop) and its association with increased blood IL6 levels at ICU admission, indicating a functional outcome related to the variant. Additionally, it mentions the impact of IFIH1 variants on gene expression and IFN response, linking these to the severity of COVID-19.

Assay Type: Genotyping

System: Real-time PCR Taqman assays

Readout: Genotype frequencies

Effect Direction: loss_of_function

Magnitude & Stats: Significant association with critical COVID-19 severity ($p=0.70$), no allelic difference noted.

Controls & Quality: Sequencing PCR fragments with different genotypes were used to validate the genotyping method.

Authors' Conclusion: No significant differences were noted for rs35744605; however, its association with recurrent viral infections suggests a potential role in COVID-19 severity.

Experiment 16 (Title: MDA5-dependent responses contribute to autoimmune diabetes progression and hindrance. PMID: 36512407)

supports_pathogenic

Justification of Paper Inclusion: The abstract discusses mutations in IFIH1, specifically mentioning an in-frame deletion within the ATPase site of MDA5 (Δ Hel1), and provides functional evidence that Δ Hel1 MDA5 protein had reduced ATP hydrolysis compared to wild-type MDA5. This indicates a wet-lab functional assay with an outcome related to the variant's effect on protein function.

Assay Type: Flow cytometry analysis

System: In vitro analysis of pancreatic cells from NOD, Δ Hel1, and KO female mice

Readout: Cell frequency and counts of proinflammatory macrophages

Effect Direction: loss_of_function

Magnitude & Stats: PE-PM comparisons showed significant differences, concluding that variant impact varies by context.

Controls & Quality: Comparison between uninfected and infected mice; independent replication of results

Authors' Conclusion: MDA5 mutations affect proinflammatory responses, directly impacting autoimmune processes

Experiment 17 (Title: IFIH1 and DDX58 gene variants in pediatric rheumatic diseases. PMID: 37342449) **supports_pathogenic**

Justification of Paper Inclusion: The abstract discusses several IFIH1 variants, including p.E627X, and reports elevated IFN-I scores in patients with these variants, indicating a functional consequence linked to the gene product's activity in the immune response. This suggests that the variants have an impact on the function of the IFIH1 protein, fulfilling the criteria for functional evidence.

Assay Type: Real-time PCR

System: RNA-based assay

Readout: IFN-I score

Effect Direction: loss_of_function

Magnitude & Stats: Elevated IFN-I scores in patients with variant.

Controls & Quality: Standard controls used for gene expression analysis.

Authors' Conclusion: Presence of variants associated with hyperactivation of IFN-I signaling pathway.

Experiment 18 (Title: IFN-I Score and Rare Genetic Variants in Children with Systemic Lupus Erythematosus. PMID: 38927451)

supports_pathogenic

Justification of Paper Inclusion: The abstract mentions rare genetic variants associated with SLE, including specific variant details for IFIH1, and indicates that the majority of these variants are functionally related to nucleic acid sensing and IFN-signaling, suggesting a wet-lab functional assay context.

Assay Type: Whole-exome sequencing (WES)

System: Clinical sequencing

Readout: Identification of genetic variants

Effect Direction: loss_of_function

Magnitude & Stats: Identified variant likely associated with SLE, functionally relevant to nucleic acid sensing

Controls & Quality: Validation according to ACMG criteria and ClinVar data

Authors' Conclusion: Variant c.1879G>T (p.E627X) found in 2 patients; associated with SLE pathogenesis.

4. ACMG/AMP Assessment

Decision: **PS3**

Strength: STRONG

Confidence: high

The majority of functional studies consistently demonstrate a loss of function associated with the p.Glu627Ter variant in IFIH1, particularly in assays measuring IFN- β production and signaling. The evidence is supported by multiple independent studies with robust controls and statistical significance, indicating that this variant likely impairs the protein's function, consistent with a pathogenic mechanism. The overall quality of the assays and the consistency of the results across different experimental setups strengthen the classification.

Key PMIDs: 20736039, 20844740, 23441136, 24686847, 28553952, 28606988, 28973304, 29018476, 29395326, 34185153, 35394816, 36434151, 36512407, 37342449, 38927451